

Homework 3

Due Date: [Check Google Classroom](#)

These exercises will give you an opportunity to practice MIPS integer arithmetic instructions and loops. Please write modular functions and strictly follow MIPS function calling and register usage conventions. Keep in view which registers are caller-saved and which are callee-saved. *Please comment your code so that we understand your approach to solve the problems.*

Exercise 1: Leap Year

Write a MIPS assembly language function `isLeapYear` which takes one integer type parameter. The parameter represents a year in Gregorian calendar. The function returns 0 if the year is not a leap year, and 1 otherwise.

Definition of Leap Year

A year is called a leap year if it is divisible by 400. If it is not divisible by 400, then it is a leap year if it is divisible by 4 **and** not divisible by 100.

By this definition, the year 2000 was a leap year because 2000 is divisible by 400. The year 2004 was also a leap year because 2004 is divisible by 4 and not divisible by 100. The year 1900 was not a leap year because 1900 is not divisible by 400, but it is divisible by 4 and also divisible by 100. By the grace of Almighty many of you will witness the year 2100. It will not be a leap year, because 2100 is not divisible by 400, and it is divisible by 4 and also divisible by 100.

Exercise 2: Date Validation

Write a MIPS assembly language function `isValidDate` which takes one integer type parameter. The parameter is an eight-digit integer `yyyymmdd`, where `yyyy` is the year, `mm` is the two-digit month, and `dd` is the two-digit day of the month. The function returns 1 if it is a valid date, and 0 otherwise. You can easily separate various parts of a date by using integer division instruction `div`, `mfhi`, and `mflo`.

The date 20050230 is not a valid date because the second month of year does not have 30 days. A date 20061301 is also not a valid date because 13 is not a valid month number. The date 20040229 is a valid date.

Exercise 3: Age in Days

Write a MIPS assembly language function `ageInDays` which takes two integer type parameter. The first parameter is the date of birth as an eight-digit number `yyyymmdd`, where `yyyy` is the year of birth, `mm` is the two-digit month of birth, and `dd` is the two-digit day of birth. If a person was born on 9 April 2004, then his birth date would be represented by 20040409. The second parameter is the current date, in the same format `yyyymmdd`. The function returns the age of the person as the number of days. This exercise will give you an opportunity to construct counting loops.

Submission Instructions

Submit the following files.

1. `<rollnumber>-h3-ex1.asm` containing solution of exercise 1
2. `<rollnumber>-h3-ex2.asm` containing solution of exercise 2
3. `<rollnumber>-h3-ex3.asm` containing solution of exercise 3

You are allowed to use the code written for an exercise in another exercise. However, you must submit three separate files so that we can easily mark your submission. **Strictly follow the above naming convention.** Your submission shall not be marked otherwise.